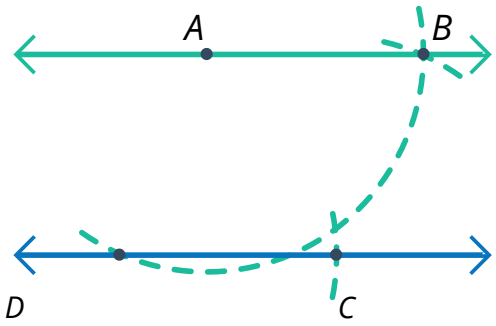
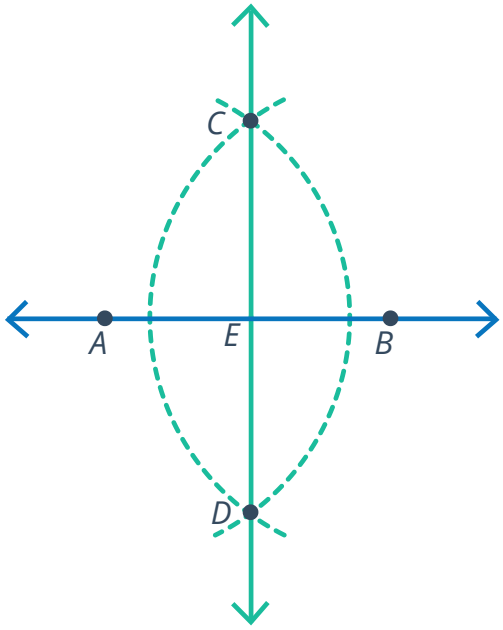
Justifying constructions using the properties of quadrilaterals

1 Consider the following diagrams. Justify the steps of the construction in a two-column proof.

a The line $\overleftrightarrow{AB}$ is constructed parallel to the given line $\overleftrightarrow{DC}$.

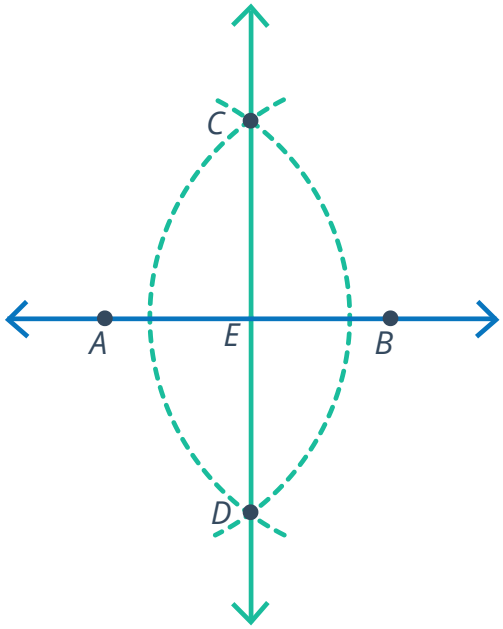


b The line $\overleftrightarrow{CD}$ is constructed so that $\overline{AB} \perp \overline{CD}$.

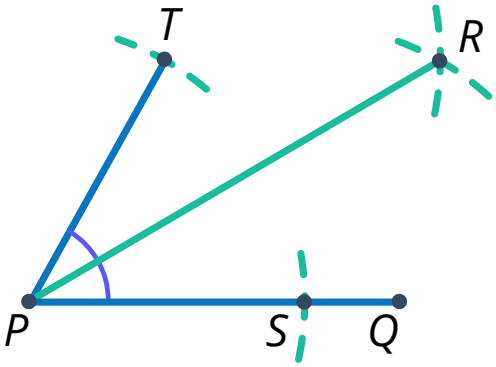


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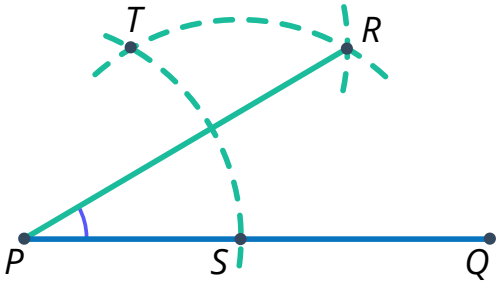
c The line $\overleftrightarrow{CD}$ is constructed so that $\overline{AE} \cong \overline{BE}$.



d $\overline{PR}$ is constructed so that $\angle TPR \cong \angle RPQ$.

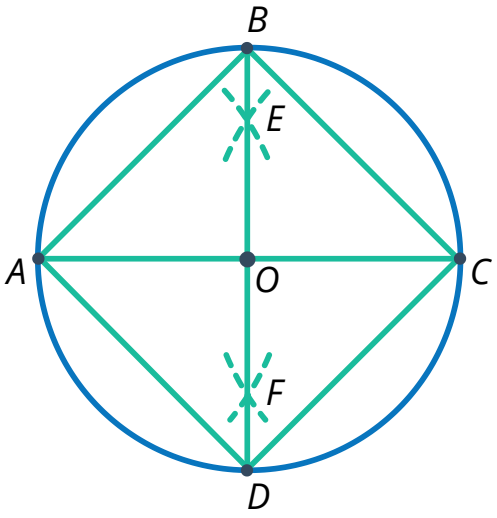


e $\overline{PR}$ is constructed from the given line segment $\overline{PQ}$ such that $m\angle RPS = 30°$.



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2 The inscribed square $ABCD$ is constructed from a given circle with a center at the point O. Justify the steps of the construction in a paragraph proof.



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